A Capacity-Radius List-Decoding Obstruction for the m=1 Specialization of Re

Author: Jerod Harbor

Status: PRE-SUBMISSION DRAFT

Front Matter

This draft claims a partial result: resolution of the $m=1$ specialization of the grand Reed-Solomon list-decoding challenge at the capacity-radius surface. It does not claim to resolve the full mutual correlated agreement challenge or the full grand list-decoding challenge for arbitrary constant m.

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Abstract

We give a capacity-radius list-decoding obstruction for the $m=1$ specialization of Reed-Solomon codes. Let $C = RS[F_q,L,k]$ with $|L|=n$. We prove that, whenever $q > \binom{n}{k+1}$, there exists a received word y for which the Hamming ball of radius $n-k$ contains exactly $\binom{n}{k}$ Reed-Solomon codewords. Consequently, if $\binom{n}{k+1} < q < 2^{128} \binom{n}{k}$, then $|\Lambda(C,y,n-k)| = \binom{n}{k} > 2^{-128} |F_q|$. Thus the $\epsilon_* = 2^{-128}$ threshold fails at $\delta = 1-k/n$ for the $m=1$ specialization in this field-size window.

1. Scope of the Claim

This draft addresses the $m=1$ specialization of the grand list-decoding challenge. In this specialization, $\Lambda(C,\delta)$ is the ordinary Reed-Solomon list at relative Hamming radius δ. The draft does not assert a result for arbitrary constant m unless an explicit m-wise extension is added.

2. Notation

Let F_q be a finite field and let $L \subset F_q$ be an evaluation domain of size n. Let $RS[F_q,L,k]$ denote the Reed-Solomon code obtained by evaluating polynomials of degree less than k on L. The rate is $\rho = k/n$. For a received word y in F_q^n and integer radius r, let $\Lambda(C,y,r)$ be the set of codewords of C within Hamming distance r from y.

3. Main Theorem

Let $C = RS[F_q,L,k]$ with $|L|=n$. Suppose $q > \binom{n}{k+1}$. Then there exists y in F_q^n such that $|\Lambda(C,y,n-k)| = \binom{n}{k}$. Consequently, if $q < 2^{128} \binom{n}{k}$, then $|\Lambda(C,y,n-k)| > 2^{-128} |F_q|$ at $\delta = 1-k/n$.

4. Proof

Choose y in F_q^n such that no polynomial of degree less than k agrees with y on more than k coordinates. Section 5 proves that such y exists when q > binom(n,k+1).

For each k-subset S of the n coordinate positions, there is a unique degree < k polynomial f_S interpolating y on S. Its Reed-Solomon codeword agrees with y on S, hence has Hamming distance at most n-k from y.

If two distinct k-subsets S and T produced the same polynomial, then that polynomial would agree with y on S union T. Since S and T are distinct k-subsets, S union T has more than k coordinates. This contradicts the choice of y. Therefore the binom(n,k) k-subsets produce binom(n,k) distinct codewords in Lambda(C,y,n-k).

Conversely, any codeword in Lambda(C,y,n-k) agrees with y on at least k coordinates. By construction, no degree < k polynomial agrees with y on more than k coordinates. Therefore each listed codeword agrees on exactly k coordinates and is generated by one of the k-subsets above. Hence |Lambda(C,y,n-k)| = binom(n,k).

If q < 2^128 binom(n,k), then binom(n,k) > 2^-128 q = 2^-128 |F_q|. Thus the threshold |Lambda| <= 2^-128 |F_q| fails at delta = 1-k/n.

5. Existence Lemma

Choose y uniformly at random from F_q^n. Fix a degree < k polynomial f and a fixed (k+1)-subset T of coordinates. The probability that f agrees with y on every coordinate of T is q^{-(k+1)}. There are q^k degree < k polynomials and binom(n,k+1) choices for T. By the union bound, the probability that some degree < k polynomial agrees with y on more than k coordinates is at most binom(n,k+1)/q. If q > binom(n,k+1), this probability is less than 1, so a suitable y exists.

6. Fixed-Rate Examples

rho = 1/2

- n = 32
- k = 16
- delta = 16/32
- radius = 16
- exact list size = 601080390
- existence threshold: q > 565722720
- threshold failure condition: q < 2^128 * 601080390

rho = 1/4

- n = 32
- k = 8
- delta = 24/32

- radius = 24
- exact list size = 10518300
- existence threshold: q > 28048800
- threshold failure condition: q < 2^128 * 10518300

rho = 1/8

- n = 32
- k = 4
- delta = 28/32
- radius = 28
- exact list size = 35960
- existence threshold: q > 201376
- threshold failure condition: q < 2^128 * 35960

rho = 1/16

- n = 32
- k = 2
- delta = 30/32
- radius = 30
- exact list size = 496
- existence threshold: q > 4960
- threshold failure condition: q < 2^128 * 496

7. Prior-Work Positioning

TO_BE_EXPANDED. The final version should compare this result with the Proximity Prize companion paper, classical Reed-Solomon list decoding, Johnson-type bounds, capacity-radius behavior, and known impossibility or near-capacity results. This section must identify whether the m=1 specialization is explicitly covered by the prize definitions or whether it is submitted as a specialization/partial result.

8. Submission Readiness

This document is a pre-submission draft until the following are added: final PDF, public repository timestamp, peer-review acceptance information, expanded prior-work comparison, and final conflict disclosure.